

Answers

3. We have U is Artinian but not Noetherian.

4. As

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \{\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \mid \varphi(1) \equiv_n 0\},$$

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/30\mathbb{Z}, \mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}.$$

5. The invariant factors are $1, 1, (1-x)^2, (1-x)^2$.

6. We have $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$.

On the other hand, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$.